

Technical Dynamics Laboratory

Manual

ME – 374 B

B.E. IIIrd Year (VIth Semester)

Contact hour per week is two

Credit – 1



**Department of Mechanical Engineering
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Department of Mechanical Engineering

B.E. Semester – VI

Technical Dynamics Lab (ME-374B)

List of Experiments

S.No.	Title	Date of Performance	Sign.	Remarks
1.	To study gyroscope and establish a relationship between torque and precession angular speed (ω_p).			
2.	To find the natural frequency of torsional vibrations also find the frequency of forced damped vibration.			
3.	To find the critical speed of shaft whirling of shaft under various type of loading.			
4.	To study the “6” Schlieren system.			
5.	To plot pressure distribution curve around a given cylinder.			
6.	To study jump phenomenon in the Cam.			
7.	Balancing of rotating masses			

Experiment 1.

Object: To study gyroscope and establish a relationship between torque and precession angular speed (ω_p).

Theory: A disc is spinning in a vertical plane parallel to plane YOZ with an angular velocity ω radians per second over the axis OX, which at the same time rotates in horizontal plane XOZ known as plane of precession, with an angular velocity ω_p radians per second as shown in **Fig. 1.1**. It is found that gyroscopic couple of magnitude T lies in the plain XOY, given by

$$T = I_p \omega \omega_p \quad (1)$$

Where, $\omega = 2\pi N/60$, N = rpm of spinning disc

t_p = Time in seconds for making one revolution of the loading arm

I_p = Polar moment of inertia of moving mass

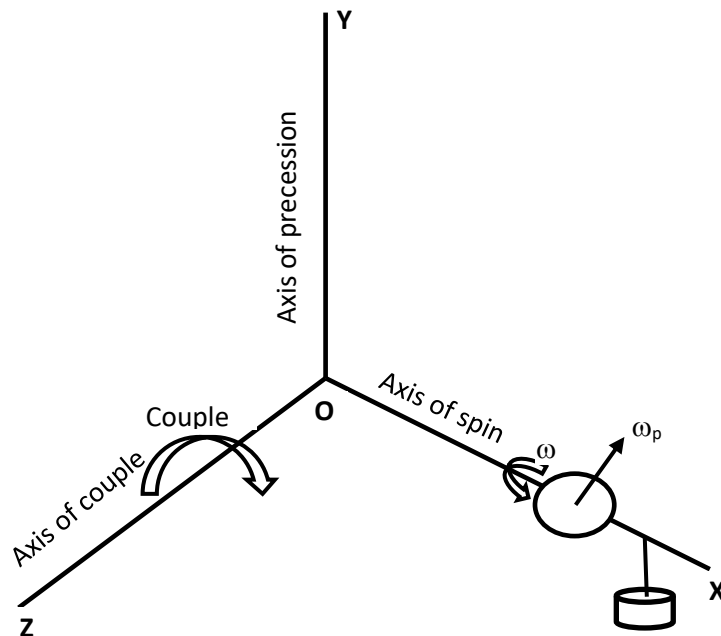


Fig. 1.1 Principle of gyroscope

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Apparatus:

The apparatus, **Fig. 1.2**. Consists of a 1/6 horse power, A.C. electric motor, capable of speed 8500 rpm. A disc flywheel of approximate polar moment of inertia $0.0905 \text{ lbs} \times \text{ft}^2$ is mounted co-axial to the armature. The motor is supported within a ring mounted on ball bearings, which is carried on a cradle attached to a vertical shaft fitted with ball bearings. Two brass slip rings are mounted on the shaft and a copper housing protects the brushes. The housing is split and can easily remove for the brushes.

A disc flywheel of approximate polar moment of inertia 38.99 g/cm^2 is mounted co-axially to the armature. A loading arm two carrying a counterpoise and hanger axis is 0.3048 m , so that when balance is obtained any load applied to hanger is $1/20$, which gives a gyroscopic torque, T equivalent to 0.3048 times load applied.

The weight of hanger is 22.68 g and slot weights supplied are:

1 number, 907.2 g C.I.

2 numbers, 453.6 g brass and

3 numbers, 226.8 g brass

The heavy base 3 of M.S. has a vertical shaft. It has four levelling screws and spirit mounted to the base for levelling. It carries a junction box, a switch and variable resistance.

A brass angular scale 5 is fitted to the cradle enable the angle of the tilt of loading arm to be found when the precession is arrested by stopping the rotation of the cradle. Knowing moment of inertia, I of parts carried on the horizontal axis, the angular velocity of tilt can be calculated.

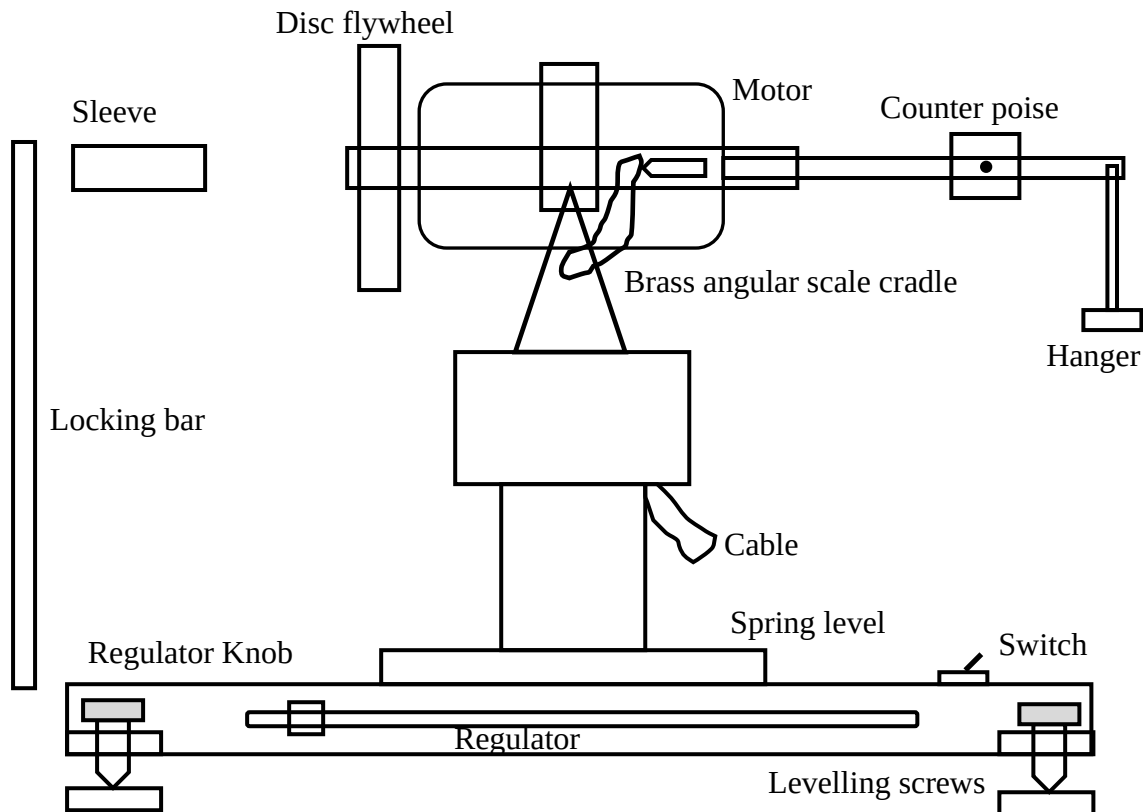


Fig. 1.2. Gyroscope

Procedure:

1. Set instrument perfectly horizontal by four travelling screws and spirit level.
2. Remove bar with two thumb screws if fitted to keep instrument perfectly free from and constraints.
3. Switch on motor increase speed up to derived value by rheostat. Wait till speed is constant.
4. Properly set the range of speedometer. Rotate the disc.
5. Move counterpoise to keep loading arm horizontal.
6. Put hanger and known weight at end of loading arm.
7. Note time for defined angular momentum of inertia of arm and find ω_p .
8. Increase load, thus torque will increase and find out corresponding ω_p .
9. Plot graph between I_p verses ω_p for constant angular velocity ω , ω_p , I_p and T of wheel and rotor can be found $T \propto \omega_p$.

Observations:

Moment of inertia of disc = 0.0905 lbs-ft²

Length of axis = 1 feet

Weight of hanger = 0.05 lbs

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$$1 \text{ lbs} = 0.45349 \text{ kg}$$

$$g = 32.18 \text{ ft/sec}^2$$

Observation Table:

S. No.	N (rpm) of spin	Time t_p (sec)	Angular velocity ω (rad/sec) = $2\pi N/60$	Angular velocity of precession $\omega_p = \theta \text{ rad}/t_p$	Weight		Torque $T_1 = I\omega\omega_p$	Torque T_2 precession (Mg.l)
					lbs	kg		

Result: As shown in the graph, the torque T_2 is proportional to the precession angular velocity (ω_p).

Experiment 2.

Object: To find the natural frequency of torsional vibrations and also find the frequency of forced damped vibration.

Theory: This experiment consists of a long elastic shaft gripped at the upper end by chuck in the bracket. The bracket is clamped by hinge support to the wall as shown in Fig. 2.1. A heavy steel flywheel clamped at the lower end of the shaft suspended from bracket. Oil pot is placed on the small table. Flywheel or disc is immersed in water / oil, which provides damping. Disc / rotor / flywheel is rotated by small angle and measure the time period of first ten cycles through stop watch.

Torsional vibrations: When shaft is twisted and untwisted alternatively torsional shear stress are induced, vibrations is known as torsional vibration.

Free natural vibrations: Elastic vibrations in which there is no friction and external force after initial perturbation of body are known as natural vibrations. Natural frequency of torsional vibration is given by:

$$\omega_n = \sqrt{\frac{q}{I}}$$

Where, ω_n = Natural frequency of torsional oscillations in rad./sec.

q = Torsional stiffness of the wire.

I = Moment of Inertia of the rotor = 0.123 kg-m²

$I = \frac{W}{g} K^2$, K = Radius of gyration of rotor.

Torsional stiffness: q is defined as the torque required on a flywheel (rotor) to produce unit angular displacement.

$$\text{i.e. } q = \frac{T}{\theta} = \frac{GJ}{L}$$

Here, G = modulus of rigidity of wire material, J = polar moment of Inertia for wire section (

$$J = \frac{\pi d^4}{32}) \text{ and } L = \text{length of wire.}$$

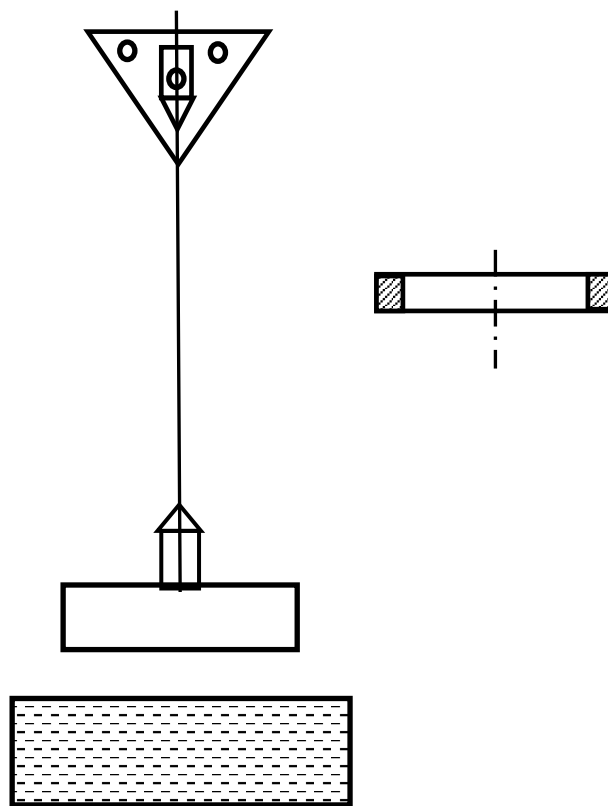


Fig. 2.1. Torsional Vibration Apparatus

Procedure: of determining Natural Frequency of Torsional vibration.

1. Give angular displacement to the flywheel (rotor or disc)
2. Record time for a few oscillations (say 10 cycles)
3. Find the time period T_1 which is the time of one cycle ($T_1 = \frac{T}{N}$), N = number of cycles.

Then, $\omega_n = \frac{2\pi}{T_1}$

4. Repeat it 3 to 4 times and find the average f .
5. Compare this value of ω_n with the ω_n theoretical.

Free damped vibrations:

The frequency of free damped vibrations is given by $\omega_{nd} = \omega\sqrt{1-\zeta^2}$

Where, ζ = damping factor of the viscous medium in which the rotor is immersed.

Procedure to determine ζ :

1. Give initial angular displacement to the flywheel (rotor or disc)
2. Note down the amplitude of two consecutive oscillations, say Y_1 , Y_2 .

Then, $\delta = \log_e \frac{Y_1}{Y_2} = \frac{2\pi\zeta}{\sqrt{1-\zeta^2}}$

δ = is known as the Logarithmic decrement.

3. Repeat it 3 to 4 times and find the average δ .

Method of determine the Natural Frequency of Damped Torsional Vibrations.

1. Give angular displacement to the rotor or disc and note down time for 10 cycles.
2. Find the time period T_2 which is the time of one cycle ($T_2 = \frac{T}{N}$), N = number of cycles.

Then, $\omega_{nd} = \frac{2\pi}{T_2}$

3. Repeat it 3 to 4 times and find the average ω_{nd} .
4. Compare this value of ω_{nd} with the ω_{nd} theoretical.

Observation Table

S.No.	Time period for 10 cycles (second)	Time period for 1 cycle (second)	$\omega_n = \frac{2\pi}{T}$ (rad/second)	Mean ω_n (rad/second)
1.				
2.				
3.				
4.				

When disc is placed in oil

S.No.	Time period for 10 cycles (second)	Time period for 1 cycle (second)	$\omega_{nd} = \frac{2\pi}{T}$ (rad/second)	Mean ω_{nd} (rad/second)
1.				
2.				
3.				

Result:

Experimental No. 3.

Object: To find the critical speed of shaft whirling of shaft under various type of loading.

Theory: On this apparatus, experiments on critical speeds of the shaft can be conducted when the shaft is supported in two, three or four bearings and loaded with discs which can be placed in any position along the shaft.

The apparatus consists of a heavy base board on which is mounted a driving motor, a variable resistance, and a switch the head stock, and brackets to which rails carry four housings. The guard housing restricts any excessive whipping of the shaft above the critical speed. The shaft is locked in self-aligning bearings by means of screw adaptors whose complete sets are supplied for each shaft. There is a three-speed drive from the motor to the head stock spindle and the latter drives the shaft through a very flexible coupling. The range of speed is from 200 to 7000 rpm. The speed at which whirling takes place is obtained by applying a tachometer to the centre in the head stock spindle.

Four test shafts are supplied viz. three 137 cm long 0.477 cm, 0.635 cm and 0795 cm diameter, respectively. Six balanced discs, three each of equal weight with the necessary adapter sleeves are also supplied. One 81.28 cm long shaft is also provided which is of 0.635 cm diameter.

Procedure:

The screwed end of shaft is fastened in the coupling with other end in the self-aligning bearing. In case of central loading a disc weight is supplied over the shaft and kept in the middle of span with guard housing. Electric motor is switched on and its speed increased slowly. The shaft rotates making a loop. Its speed is further increased to say four times it vibrates with a vote at

the centre and at each support forming loops – speed is again measured the speed at which the amplitude of loop is maximum, is the critical speed.

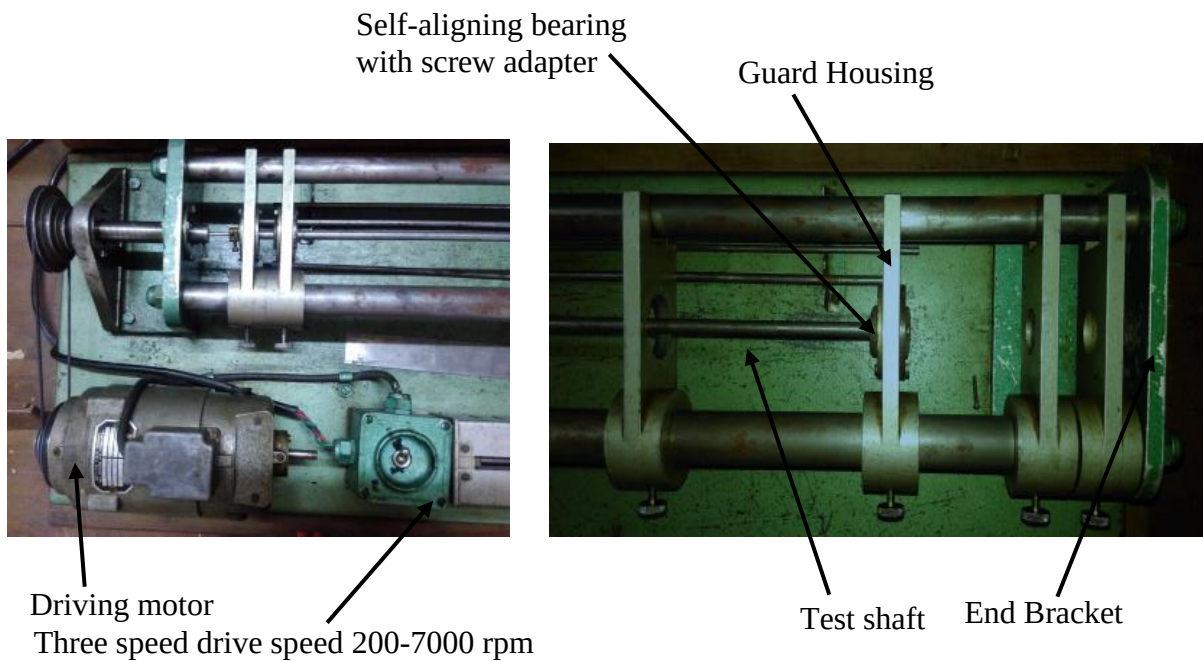


Fig. 3.1 Whirling of shaft apparatus

Formula:

1. Critical speed when shaft is between two bearings with no overhang.

$$\delta_s = \frac{5}{384} \times \frac{WL^3}{EI} \text{ in meter}$$

Where, W = weight of shaft, L = Length of span, E = Modulus of elasticity and I = Moment of

Inertia of shaft = $\frac{\pi}{64} d^4$

$$\text{Critical speed } N_c = \frac{0.5615}{\sqrt{\delta_s}} \text{ Hz}$$

2. Critical speed of shaft with concentrated load at mid span.

$$\delta = \frac{1}{48} \times \frac{WL^3}{EI} \text{ in meter}$$

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$$\text{Critical speed } N_c = \frac{0.4985}{\sqrt{\frac{\delta_s}{1.27} + \delta}} \text{ Hz}$$

Calculations:

$$\delta_s = \frac{5}{384} \frac{WL^3}{EI} \quad (\text{for simply supported beam UDL beam})$$

$$W = \rho v g$$

$$W = A \times L \times \rho \times g = \frac{\pi}{4} D^2 \times l \times \rho \times g = 1.890 \text{ kg}$$

$$d = 0.0063 \text{ m}, \rho = 7830 \text{ kg/m}^3 \text{ and } l = 0.790 \text{ m}$$

$$I = \frac{\pi}{64} d^4 = 7.728 \times 10^{-11} \text{ m}^4$$

$$E = 2.1 \times 10^{11} \text{ N/m}^2$$

$$\delta_s = 7.4764 \times 10^{-4} \text{ m}$$

$$N_c = \frac{0.5615}{\sqrt{\delta_s}} = 20.535 \text{ rps} = 1232.12 \text{ rpm}$$

Observation table:

S. No.	Diameter of shaft (mm)	Span length (mm)	Concentric load (kg)	Deflection self-weight	Critical	Speed of shaft
1.						
2.						
3.						

Result:

1. When the load on shaft increased then critical speed of shaft decreases.
2. When diameter of shaft measured than critical speed decreased.

Experiment No. 4

Object: To study the “6” Schlieren system.

Apparatus: “6” Schlieren system consists of 6” die and 82” focus length concave mirror complete is mount and cover 2 NOS.

- Optical bench $\frac{1}{2}$ meter long – 2 NOS
- Carriage – 6 NOS
- Condenser lens 3” diameter on stand – 1 No.
- Lamp housing on stand with stand – 1 No.
- Adjustable knife – 1 No.
- Image focusing lens and mirror on stand with optics
- Adjustable slit 1 No.
- Ground glass screen in frame – 1 No.

Instruction: The function of the schlieren system is determine the small variation in the index of refraction of transparent material. It can be shown that to a rough approximate the schlieren method to measure the first deviate of refraction index.

The basic idea of this method was originated by fault for testing lenses and other optical parts but toppler greatly developed this method and applied to study the various air disturbances produced by sparks explosions, air flows etc. for the study of the rapidly moving devised a spark light sources which could be timed to illuminated a moving air distributed desired instant. As a spark light source which could be timed to illuminated a moving air disturbance desired instant as a result snapshot picture of the order after microseconds duration could be taken of the phenomenon schlieren method has been applied also due to visualisation and study of turbulence and connections due to hot surface the air flow around projections fit widely used in investigation of the air flow around bodies placed in high wind tunnels.

Description: The general principle of Schlieren is illustrated by Figure which is essentially the arrangement used by toppler, wood and other light from a source. A such as a straight or ribbon filament perpendicular to the page is focused by the lens B on the slit. So that it is uniformly distributed the light passing, S by long focus large diameter so called

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schlieren lens S. if the knife edge S_2 is more upward the beam until it just blocks out the image of S. No light will be reach P provided the lens C is of good optical quality is free of chromotion and spherical observation Fig. 2 shows a twin mirror system that gives good resolving power of Schlieren mirror C and E are a carefully method pair usually they are made parabolic to better one tenth of a wave length of light.

The excellence of their optical quality and it is over emphasised also due to size (more than a foot in diameter) and weight they must be carefully mounted to avoid distribution the O_2 must be approx. equal to angle O_2 and they should be as small as possible. Although angles up to about 70° are usually successful. The distance between the mirror C and E is not critical but, it is good practice to make greater to twice the focal length of the mirror. Also due to the optical system.

Experiment No. 5.

Object 5. To plot pressure distribution curve around a given cylinder.

Specifications:

The section size 25 cm × 25 cm. Maximum speed at clear test section, 30 m/second. Area ratio 13. Motor hp 3 at 2800 rpm.

Description:

The wind tunnel, see Fig. 5.1 is made up of heavy gauge M.S. sheet in MS angle frame with smooth conductors for undistributed flow. The entrance area is square in section of size 90 cm × 90 cm. M.S. section on downstream side where the cross-sectional area is 25 cm × 25 cm contains lugs for mounting over M.S. angle feet. Test section has two transparent sides of 2 mm perspex sheet with aluminium beading over sturdy M.S. frame. A 15 cm × 15 cm door provision is made at the bottom for fixing models and taking out probing tubes. On the test section near the entrance a traversing mechanism for the pilot tube is provided on this door. Towards the downstream side a transition piece 25 cm × 25 cm square to 50 cm circle is designed for reducing losses and smooth flow. The next part is the control section 50 cm diameter and 92 cm long having a butterfly valve controlled by a hand lever at the front side. A calibration chart is printed against the lever for convenience to indicate the approximate air speed at test section and horse power consumed without model. The lever can be fixed in any position by screwing down the knurled unit. The last part of the wind tunnel is the axial blower section 50 cm diameter and 1 m long containing a specially designed axial flow blower with the least noise for the running speed of 2800 rpm. It is run by 3 hp, 440-volt, 50 cycle/second, 3 phase, A.C. motor. The terminal box for the motor connections is fixed at the front side of the name plate. A starter and an isolation switch are supplied with the motor. The start-delta type starter is fixed on the wall.

A 15-tube water manometer is used with the wind tunnel to measure the static pressure around the model placed in the test section. The model contains a nut at the bottom which is removed and model inserted into the test section and locked in position by tightening the nut from outside. The probe tubes coming from the various tapping on the model are then connected to the manometer. The model can be fixed in any orientation as desired.

Traversing mechanism

The traversing mechanism provided 15 cm movement in the horizontal direction and almost an equal movement in vertical direction starting from the top of the test section. In designing the traversing mechanism care has been taken to see that the test section of 25 cm × 25 cm changes by a small amount and there is no leakage of air. The pitot tube is held securely by a clamping device incorporated in traversing mechanism. The multi manometer is of tilting type and its inclination is measured from the indicating dial.

Apparatus:

The apparatus consists of wind tunnel complete with multitube manometer, micromanometer.

A paper is wrapped over the cylinder, the holes 14 in number are picked and from the development of cylinder, the angles are measured. The holes are numbered as shown in Fig. 5.2. The cylinder is placed with its axis vertical in the test section of the wind tunnel. Dynamic or velocity pressure at any point i of the cylinder where $i = 1, 2, 3, \dots, 14$ is $(p_{ti} - p_s)$ is given as follows,

$$\frac{\rho C_s^2}{2} = p_{ti} - p_s$$

Or Dynamic pressure = stagnation or total pressure – static pressure

Dynamic pressure can be found out by micromanometer by connecting the stagnation limb of the Prandtl tube to its +ive side and static limb to the negative side, for plotting purposes, $(p_{ti} - p_s)$ is made non-dimensional by dividing it by dynamic pressure of the free stream i.e.

$$\frac{p_{ti} - p_s}{\frac{1}{2} \rho C_s^2}$$

This ratio is found out for various points $i = 1, 2, 3, \dots, 14$ and can be plotted radially against the position of the points on the surface of the cylinder by selecting a suitable scale, negative ratio towards outside of the periphery and positive ratio towards inside. Figure 5.3 shows the shape of pressure distribution curve. It is found that for various stream.

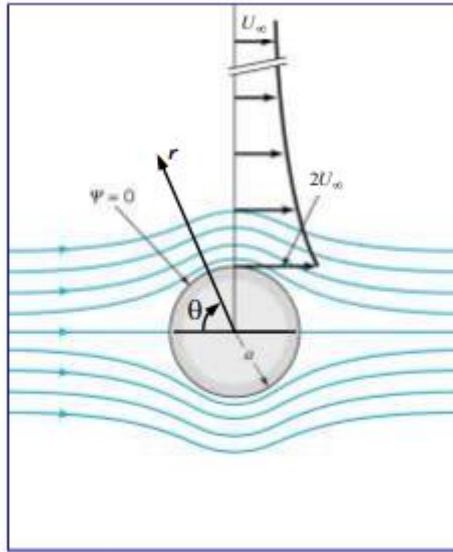
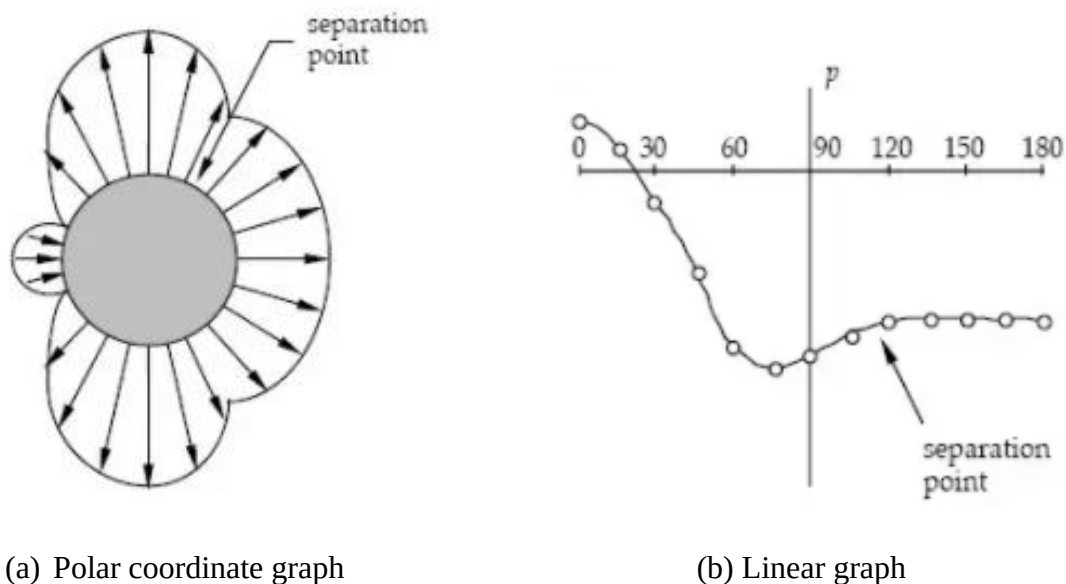


Fig. 3.1. Ideal potential flow around a circular cylinder

$$\frac{p_{ti} - p_s}{\frac{1}{2} \rho C_s^2} = 0 \text{ at about } 39 \text{ to } 40^\circ.$$

Velocities measured from the direction of flow shown by the arrow. These angles are known as critical angles. For these points $p_{ti} = p_s$.



(a) Polar coordinate graph

(b) Linear graph

Fig. 3.2 Pressure distribution around a circular cylinder placed in a uniform flow

Fluctuation is maximum at about 90° to the direction of flow. At these particular region separations of boundary layer is most likely to occur further this point of separation keeps on moving along with the circumference of the cylinder through about $30 - 40^\circ$ and fluctuation of

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manometric readings in this region. At per Binder, in the Reynold's range of 50000 to 200000, the transition in the boundary layer from laminar to turbulent occurs at the point of separation causing wide fluctuations in pressure readings in this region.

Experiment No. 6.

Object: To study the jump phenomenon in the Cam.

Description: In a cam-follower system, the contact between the cam surface and the follower is maintained by means of a retaining spring. Beyond a particular speed of cam rotation, the follower may lose contact with the cam, because of inertia force acting on the follower. This phenomenon is called as “Jump Phenomenon”. When the follower re-establishes contact with the cam, it may do so with severe impact loads that can damage the surface of the cam and “hammering nose” can be heard at this jump speed.

During the follower jump, transient vibrations are set up in the follower and these occur only with high speed, highly flexible cam-follower system. With jump, the cam and follower separate owing to excessively unbalanced forces exceeding the spring force during the period of negative acceleration. This is undesirable since the fundamental function of the cam-follower system, the constraint and control of follower motion are not maintained. Also, the life of the cam flank surface reduces due to hammering action of follower on cam and hammering noise is generated which further results in vibrations of the system. The jump phenomenon will be avoided by limiting the speed of cam or by increasing the stiffness of the retaining spring.

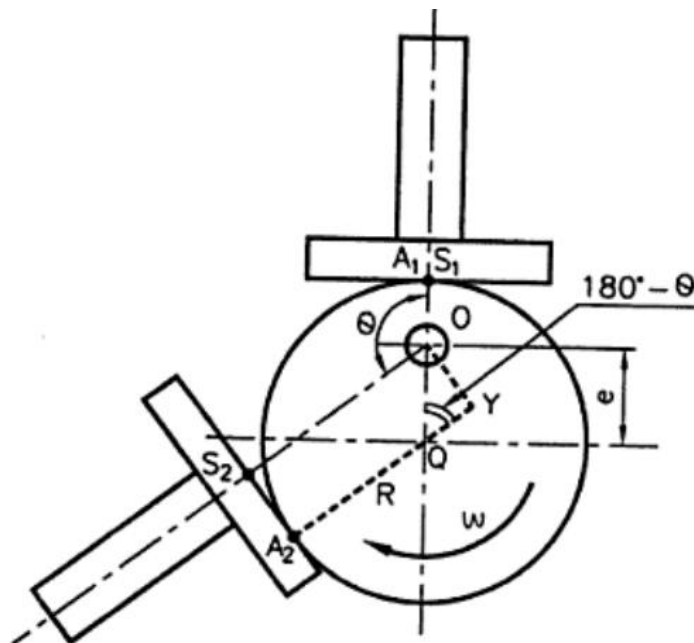


Fig. 6.1

Fig. 6.1 shows an eccentric cam follower system, which is analysed for jump phenomenon.

From Fig. 6.1, we have:

$$\text{Lift of follower} = y = OS_2 - OS_1$$

$$= A_2Q + QY - OS_1$$

$$= R + e \times \cos(180 - \theta) - (R - e)$$

$$y = e \times (1 - \cos \theta) \quad (6.1)$$

Differentiating equation (6.1) with respect to time t , we get:

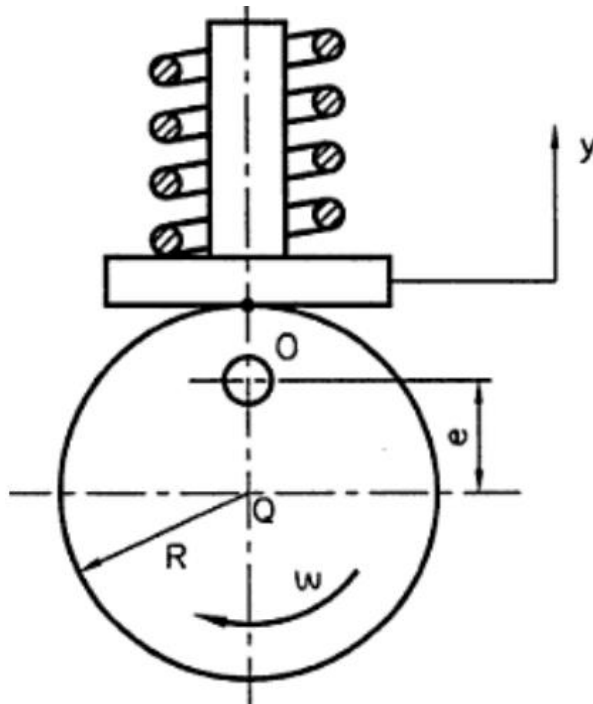
$$\text{Velocity of follower} = \dot{y} = e\omega \sin \theta \quad (6.2)$$

Differentiating equation (6.2) with respect to time t , we find:

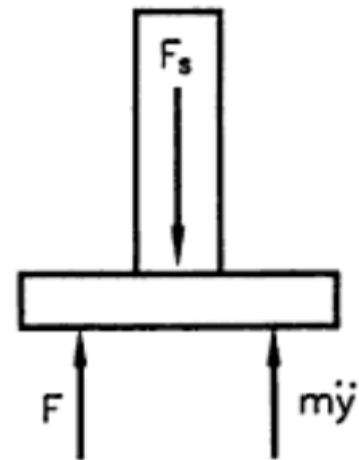
$$\text{Acceleration of follower} = \ddot{y} = e\omega^2 \cos \theta \quad (6.3)$$

$$\text{Where, } \theta = \omega.t = \text{cam angle turned from lowest position} \quad (6.4)$$

Now, consider the arrangement as shown in **Fig. 6.2**, then



(a) Actual System



(b) Free body diagram (FBD)

Fig. 6.2

By the **Fig. 6.2**, it is represented that, m = mass of follower

e = eccentricity

k = stiffness of spring

F = constant force between cam and follower

F_s = total spring force

ω = Angular speed of cam

P = preload in spring

Then, from free body diagram, we have:

$$InertiaForces = \sum ExternalForces$$

$$-m\ddot{y} = F_s - F$$

$$F = m\ddot{y} + F_s \quad (6.5a)$$

$$F_s = P + ky$$

$$F_s = P + k \times e(1 - \cos \omega t) \quad (6.5b)$$

Substituting equation (6.3) and equation (6.5b) into equation (6.4a), we get:

$$F = m \times (e\omega^2 \cos \omega t) + P + k \times e(1 - \cos \omega t)$$

$$F = (m\omega^2 - k) \times e \times \cos \omega t + (P + ke) \quad (6.6)$$

This contact force between cam and follower is maximum when $\theta = 0^\circ$ and minimum when $\theta = 180^\circ$. It is also dependent upon the square of cam velocity.

When this contact force between cam and follower becomes negative (i.e. less than zero); the follower would lose contact with the surface resulting in jump. This would happen if the speed is increased beyond a particular critical speed ' ω_j ' (at $\theta = 180^\circ$).

$$F = 0 = (m\omega_j^2 - k) \times e \times \cos 180 + (P + ke)$$

Where ω_j = jump speed

$$0 = (m\omega_j^2 - k) \times (-e) + (P + ke)$$

$$(m\omega_j^2 - k) \times (e) = (P + ke)$$

$$me\omega_j^2 = P + 2ke$$

$$\omega_j = \sqrt{\frac{P + 2ke}{me}}$$

Therefore, to avoid jump $\omega_j < \sqrt{\frac{P + 2ke}{me}}$

To avoid jump: $P > (m\omega^2 - 2k)e$

Experiment No. 7.

Object: Static and dynamic balancing of rotating masses.

Description: Shafts which revolve at high speeds must be carefully balanced if they are not to be a source of vibration. If the shaft is only just out of balance and revolves slowly. The vibration may be a nuisance but catastrophic failure can occur at high speeds even if the imbalance is small.

For example, if the front wheel of a car is slightly out of balance this may be felt as a vibration of the steering wheel. However, if the wheel is seriously out of balance, control of the car may be difficult and the wheel bearings and suspension will wear rapidly, especially if the frequency of vibration coincides with any of the natural frequencies of the system. These problems can be avoided if a small mass is placed at a carefully determined point on the wheel rim.

It is even more important to ensure that the shaft and rotors of gas turbine engines are very accurately balanced. since they may rotate at speeds between 15000 and 50000 rev/min. At such speeds even slight imbalance can cause vibration and rapid deterioration of the bearings leading to catastrophic failure of the engine.

It is not enough to place the balancing mass such that the shaft will remain in any stationary position, that is in static balance. When the shaft rotates, periodic centrifugal forces may be developed which give rise to vibration. The shaft has to be balanced both statically and dynamically.

Usually, shafts are balanced on a machine which tells the operator exactly where he should either place a balancing mass or remove material. The TM102 apparatus requires the student to balance a shaft by calculation or by using a graphical technique, and then to assess the accuracy of his results by setting up and running a motor driven shaft. The shaft is deliberately made out of balance by clamping four blocks to it, the student being required to find the positions of the third and/or fourth blocks necessary to statically and dynamically balance the shaft.

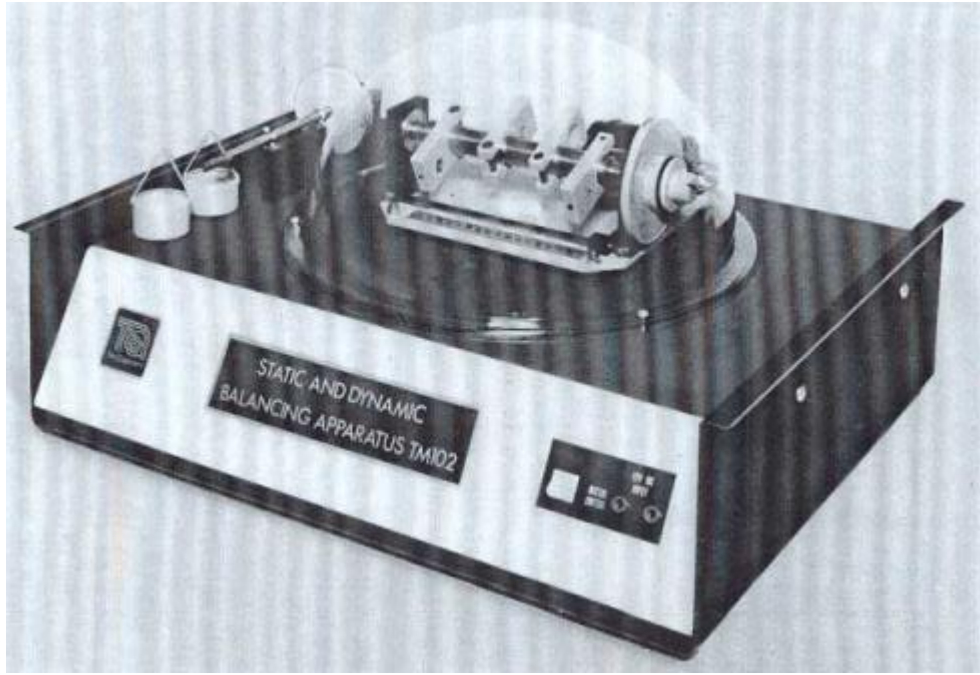


Fig. 7.1 Static and Dynamic Balancing Apparatus

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A shaft with masses mounted on it can be both statically and dynamically balanced. If it is statically balanced, it will stay in any angular position without rotating. If it is dynamically balanced, it can be rotated at any speed without vibration. It will be shown that if a shaft is dynamically balanced it is automatically in static balance, but the reverse is not necessarily true.

Static Balance

Figure 7.1 shows a simple situation where two masses are mounted on a shaft. If the shaft is to be statically balanced, the moment due to weight of mass (1) tending to rotate the shaft clockwise must equal that of mass (2) trying to turn the shaft in the opposite direction. Hence for static balance,

$$W_1 r_1 = W_2 r_2 \quad (7.1)$$

The same principle holds if there are more than two masses mounted on the shaft, as shown in Figure 7.2. The moments tending to turn the shaft due to the out of balance masses are:

Mass	Moment	Direction
(1)	$W_1 r_1 \cos \alpha_1$	Anti-clockwise

(2)	$W_2 r_2 \cos \alpha_2$	Clockwise
(3)	$W_3 r_3 \cos \alpha_3$	Clockwise

For static balance,

$$W_1 r_1 \cos \alpha_1 = W_2 r_2 \cos \alpha_2 + W_3 r_3 \cos \alpha_3 \quad (7.2)$$

Anti-clockwise Clockwise

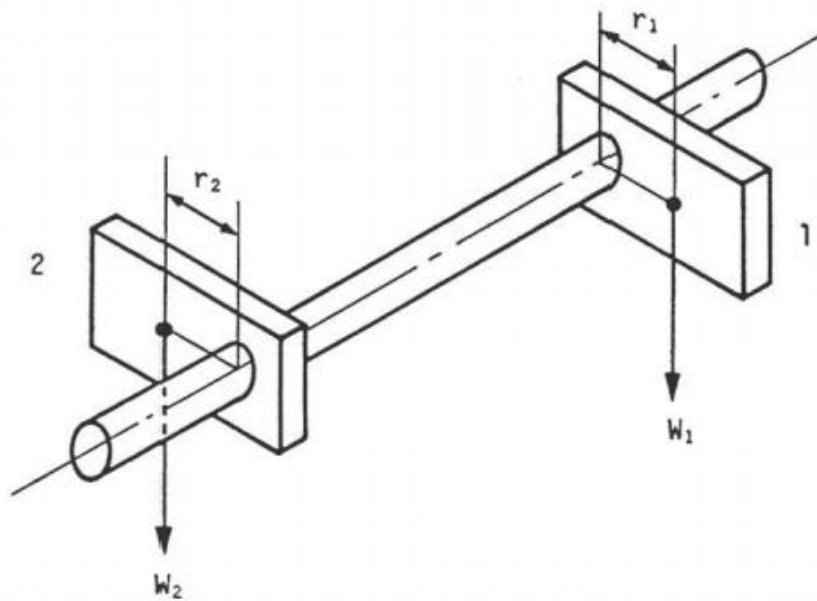


Fig. 7.2 Simple Two Mass System

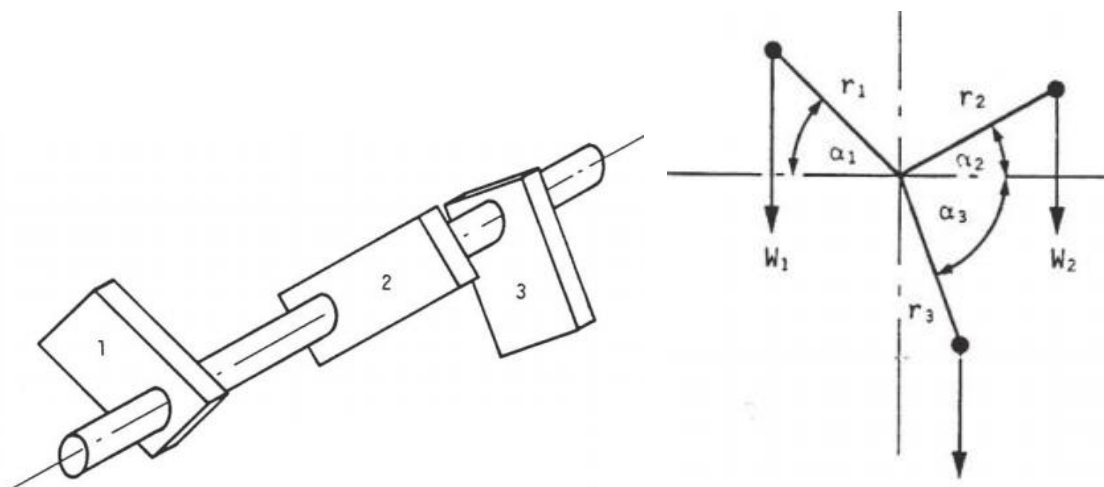
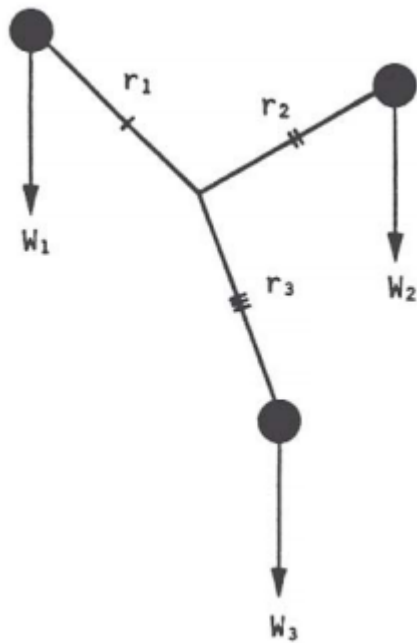
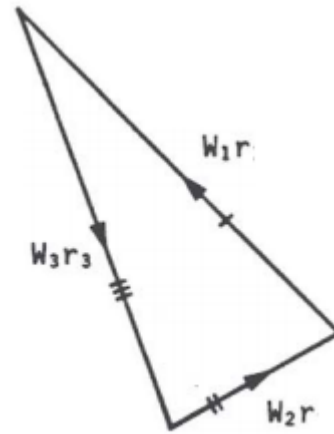


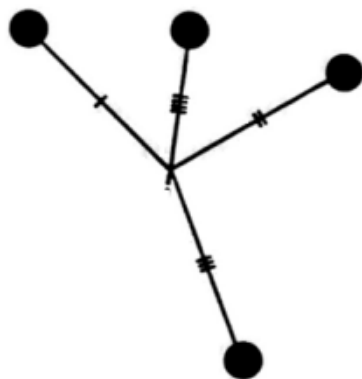
Fig. 7.3 Three Mass System (System representations of shaft and masses)



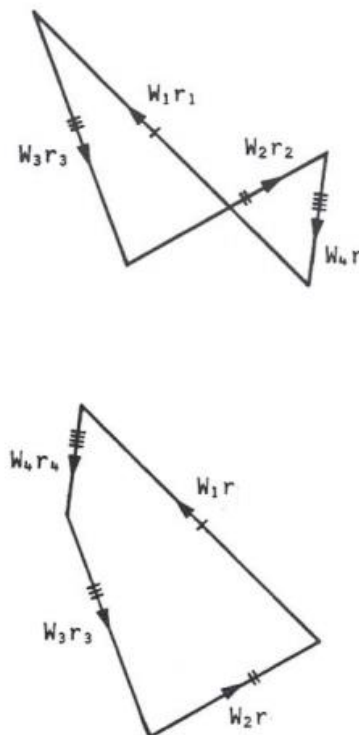
(a) Three mass system



(b) Moment Triangle



(c) Four mass system



(d) Moment Triangle

Fig. 7.4 Moment Polygons for Static Balance of 3 and 4 Mass Systems

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In general, values of W , r and a have to be chosen such that the shaft is in balance. However, for experiments using the TM102 apparatus, the product Wr can be measured directly for each mass and only the angular positions have to be determined for static balance.

If the angular positions of two of the masses are fixed, the position of the third can be found either by trigonometry or by drawing. The latter technique uses the idea that moments can be represented by vectors as shown in figure 2.4(a). The moment vector has a length proportional to the product Wr and is drawn parallel to the direction of the weight from the centre of rotation.

For static balance the triangle of moments must close and the direction of the unknown moment is chosen accordingly. If there are more than three masses, the moment figure is a closed polygon as shown in figure 2.4(b). The order in which the vectors are drawn does not matter, as indicated by the two examples on the figure.

If on drawing the closing vector, its direction is opposite to the assumed position of that mass, the position of the mass must be reversed for balance. For example, mass (4) shown in figure 2.4(b) must be placed in the position shown dotted to agree with the direction of vector W_4r_4 .

Dynamic Balance:

The masses are subjected to centrifugal forces when the shaft is rotating. Two conditions must be satisfied if the shaft is not to vibrate as it rotates:

- (a) there must be no out of balance centrifugal force trying to deflect the shaft.
- (b) there must be no out of balance moment or couple trying to twist the shaft.

If these conditions are not fulfilled, the shaft is not dynamically balanced.

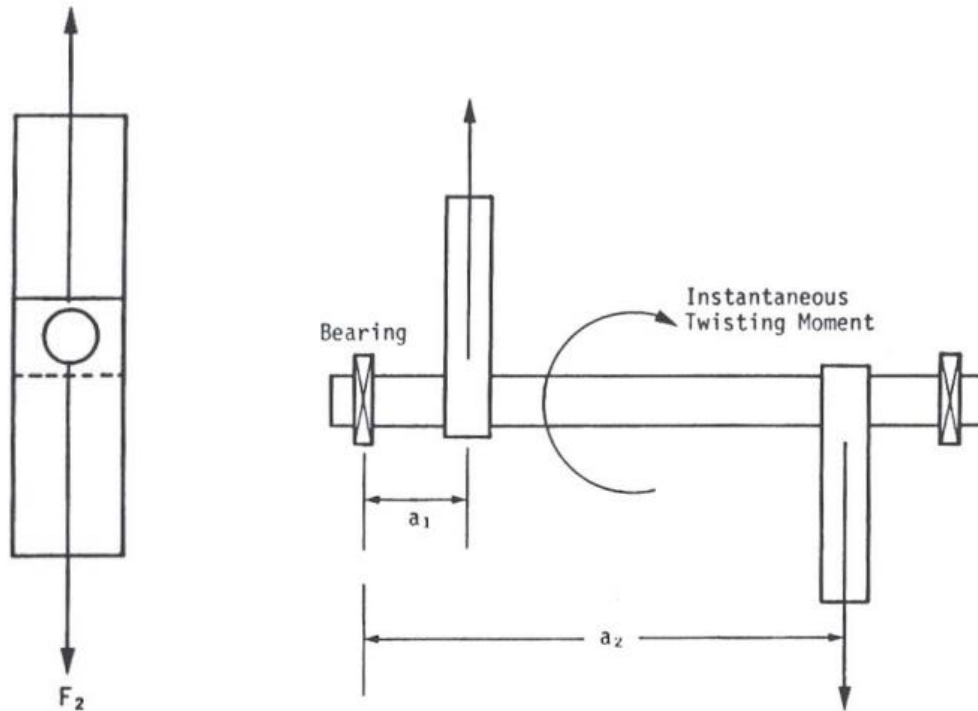


Fig. 7.5 Dynamic Out-of-Balance for two mass system

Applying condition (a) to the shaft shown in figure 7.4 gives:

$$F_1 = F_2 \quad (7.3)$$

The centrifugal force is $mr\omega^2$ or $\frac{W}{g}r\omega^2$:

$$\frac{W_1}{g}r_1\omega^2 = \frac{W_2}{g}r_2\omega^2 \quad (7.4)$$

The angular speed of rotation is the same for each mass so that for dynamic balance:

$$W_1r_1 = W_2r_2 \quad (7.5)$$

This is the same result obtained in equation 7.1 for the static balance of the shaft. Thus, if a shaft is dynamically balanced it will also be statically balanced.

The second condition is satisfied by taking moments about some convenient datum such as one of the bearings.

$$\text{Thus,} \quad a_1F_1 = a_2F_2 \quad (7.6)$$

But from equation 7.3, $F_1 = F_2$, so that $a_1 = a_2$. Thus, in this simple case, dynamic balance can only be achieved if the two masses are mounted at the same point along the shaft.

Since the reference point is immaterial, it is usually convenient to take moments about one of the masses so that the effect of that mass is deleted from the moment equation. Taking moments about mass (1) produces the result

$$F_1(a_2 - a_1) = 0 \quad (7.7)$$

Since the centrifugal force cannot be zero, a_1 must equal a_2 as proved above.

Unlike static balancing where the location of the masses along the shaft is not important, the dynamic twisting moments on the shaft have to be eliminated by placing the masses in carefully calculated positions. If a shaft is statically balanced it does not follow that it is also dynamically balanced.

Dynamic Balancing of Three Masses

It has been shown that dynamic balance can only be achieved for a two-mass system if the masses are mounted at the same point along the shaft. It can also be shown that there are special conditions which must be satisfied for dynamic balancing of three masses.

Consider the case shown in figure 7.5(a). Mass (3) is positioned vertically for convenience. Condition (b) for dynamic balance can be expressed mathematically by equating moments for centrifugal forces in both horizontal and vertical planes. In order to simplify the equations it is convenient to take moments about mass (1) so that moments due to forces on this mass are eliminated.

$$\text{Horizontal: } a_2 F_2 \cos \alpha_2 = 0$$

$$\text{Vertical: } a_2 F_2 \sin \alpha_2 = a_3 F_3 \quad (7.8)$$

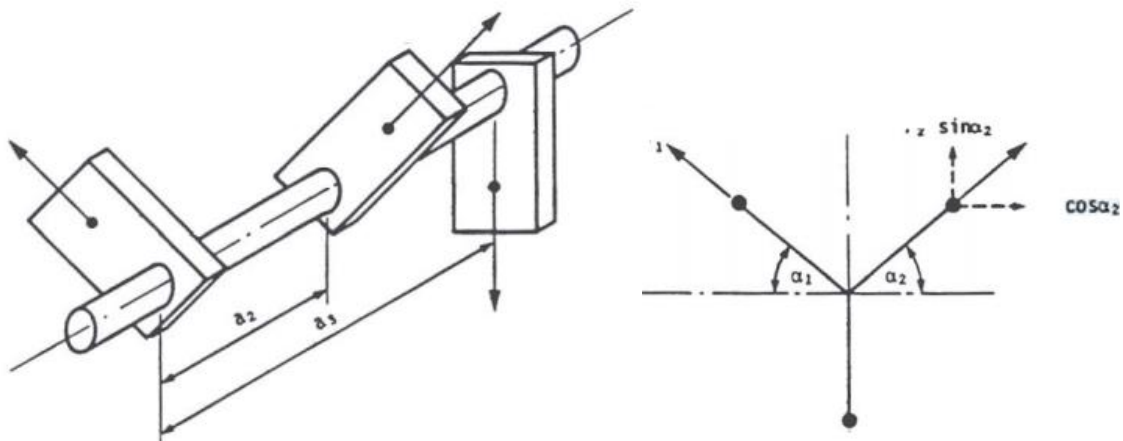
The conditions which satisfy equation 2.7 are that either $a_2 = 0$ or $a_2 = 90$ or 270° (i.e. $\cos \alpha_2 = 0$). Substituting these into equation 2.8 gives the following conditions:

Case 1: $a_2 = 0$

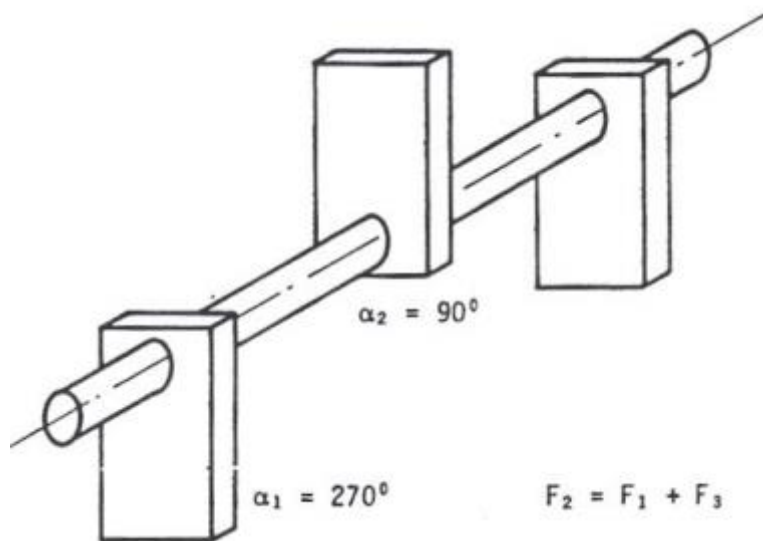
For this condition a_3 must also be zero. Thus, for arbitrary values of a_2 and a_3 , all three masses must be located at the same point along the shaft.

Case 2: $\alpha_2 = 90$ or 270°

For these conditions it is necessary to write down further equations to obtain solutions.



(a) General case for three mass system



(b) Actual solution for axially spaced masses

Fig. 7.6

Horizontal: $F_1 \cos \alpha_1 = F_2 \cos \alpha_2$

Vertical: $F_3 = F_1 \sin \alpha_1 + F_2 \sin \alpha_2$ (7.9)

If $\alpha_2 = 90^\circ$, equation 2.9 gives $\alpha_1 = 90$ or 270° . Assuming that $\alpha_1 = 90^\circ$, equation 7.9 reduces to:

$$F_3 = F_1 + F_2 \quad (7.10)$$

Also, equation 7.8 reduces to:

$$a_2 F_2 = a_3 F_3 \quad (7.11)$$

Combining these two equations and solving for F_1 gives:

$$F_1 = F_3 \left(-\frac{a_3}{a_2} \right) \quad (7.12)$$

Now, if a_3 is greater than a_2 (as indicated in figure 7.5) it follows that F_1 must be negative, and α_1 must be 270° and not 90° as assumed above. The resulting configuration for dynamic balance is shown in figure 7.5(b).

Thus, if the masses are distributed along the shaft, the following conditions must be satisfied for dynamic balance:

- (a) Central mass at 180° to other two masses
- (b) Masses chosen such that.

$$F_3 = F_1 + F_2 \quad (7.13)$$

- (c) Masses distributed along the shaft such that,

$$a_2 F_2 = a_3 F_3 \quad (7.14)$$

Dynamic Balancing of More Than Three Masses

If there are more than three individual masses, there are no special restrictions which apply to the angular and shaft-wise distributions of the masses and the general conditions for dynamic balance have to be applied to obtain solutions.

The angular positions of the masses can be determined by applying conditions for either static balance or for dynamic balance. The distribution of the masses along the shaft is then found by applying Condition (a) for dynamic balance. This can be done either by calculation, or by a drawing method similar to that described for static balancing.

7.2 EXPERIMENTAL APPARATUS

The TM102 Static and Dynamic Balancing Apparatus (figure 7.2.1) consists of a perfectly balanced shaft to which a set of four rectangular blocks can be clamped. The shaft runs in ball bearings and is driven by a 12 volt electric motor through a belt and pulley arrangement. The

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motor and shaft are mounted on a metal plate which is supported on resilient rubber feet. These feet allow the plate to vibrate when the shaft is out of balance.

Discs having eccentric holes of differing diameters can be fitted to the rectangular blocks so that the out of balance moment of each block can be varied. The blocks can be clamped to the shaft in any angular and shaft-wise positions. Two hexagon keys are provided with the apparatus. The larger one is used to clamp the blocks to the shaft, whilst the smaller key fits the screws retaining the eccentric discs.

Linear and protractor scales are provided for setting and measuring the positions of the blocks. Adjacent to the linear scale is a slider which can be pushed up against each block in turn to read the shaft-wise position. The slider is also used as a datum stop against which the blocks can be held when reading the angular position.

The out of balance moments are measured by first fitting the removable shaft and pulley to the main shaft. Each rectangular block is then clamped to the shaft in turn. A cord having a weight bucket on each end is hung on the extension pulley and ball bearings are placed in the bucket until the block has moved through 900. The out of balance moment is proportional to the number of balls required.

Micro switches actuated by a perspex dome ensure that the shaft cannot be driven unless the dome is in position. The motor switch on the front panel is spring loaded so that the motor cannot be left running. The motor requires a supply of 12 Volts at 2 amps which may be provided either by one of the TecOupment (i.e. E66/E90 Speed Control) or by an ordinary automotive battery.

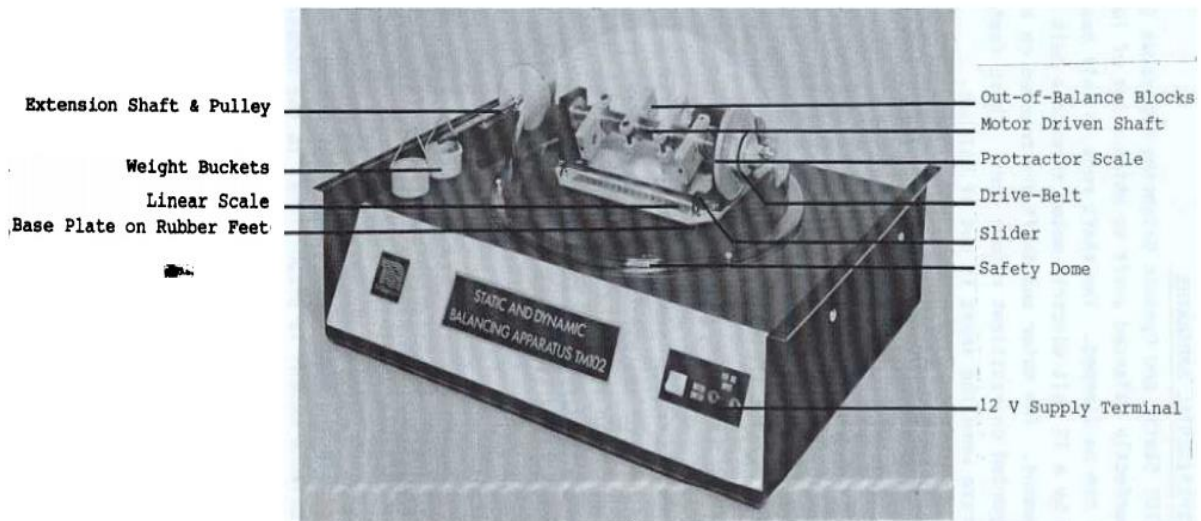


Fig. 7.2.1 Main Features of the Apparatus

EXPERIMENTAL PROCEDURE AND TYPICAL RESULTS

1. Demonstration of Static Balance but Dynamic Imbalance
2. Remove the perspex dome and shaft drive belt.
3. Remove the discs from the four rectangular blocks using the smaller hexagon key.

Set up two of the blocks as shown in figure 7.2.2 (a) with a relative angular displacement of 180° and shaft-wise displacement of 120mm. Use the slider to set and read off the positions of the blocks. The larger hexagon key fits the screws which clamp the rectangular blocks onto the shaft.

4. Observe that the shaft will remain in any angular position and is therefore statically balanced.
5. Connect the apparatus to a 12 volt supply. Make sure that the slider adjacent to the linear scale is clear of the blocks, then replace the shaft drive-belt and the perspex dome.
6. Briefly run the motor. Note the severe imbalance of the shaft.

Simple Dynamic Balance using Four Masses

1. Remove the perspex dome and set up the four rectangular blocks as shown in figure 7.2.2 (b).
2. Prove theoretically that the shaft is both statically and dynamically balanced.
3. Test the shaft for static balance.

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4. Replace the drive-belt and the perspex dome. Run the motor and observe the lack of vibration, showing that the shaft is dynamically balanced.

An alternative is to place the two middle blocks adjacent to one another such that their mating faces are mid-way between the two end blocks. This gives a configuration equivalent to the three-mass system shown in figure 7.2.3 (b) for which $F_2 = 2F_1 = 2F_3$. The special condition for dynamic balance of a 3-mass system (equation 7.13) is therefore satisfied so the shaft is in balance.

*** Always take the blocks off the shaft before removing or replacing the discs.**

Note: The motor and rotating shaft assembly is mounted on a metal plate which may be clamped down. Ensure the plate is clamped fully down for static balancing tests. For dynamic balancing release the clamps, turn through 90° and lock down, leaving the plate free to vibrate.

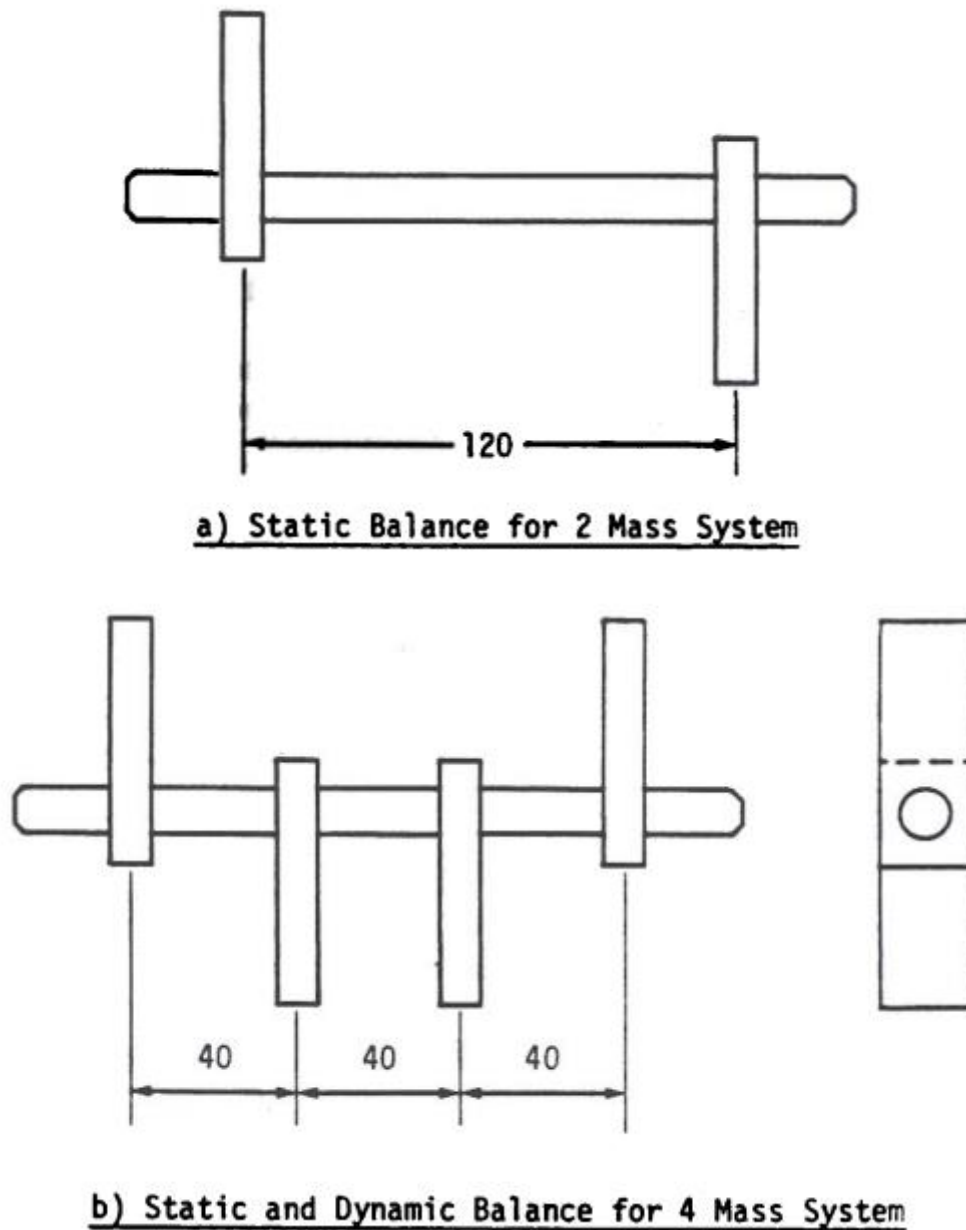


Fig. 7.2.2 Configurations for Simple Demonstrations

Static and Dynamic Balance of a Four Mass System

Experimental determination of (Wr) values

1. Remove the perspex dome and remove the shaft drive belt.
2. Unclip the extension pulley and insert it in the pulley end of the motor driven shaft.
3. Move the apparatus to the edge of the table or bench. Loop two or three turns of the weight bucket cord around the extension pulley. Ensure that there are no obstructions to the movement of the weight buckets.

4. Insert the eccentric disc with the small hole into one of the rectangular blocks. Clamp the block to the shaft such that the protractor scale reading is 0° . Call this block (1).
5. Gradually add the steel balls to one of the weight buckets until the block has moved through 90° . Whilst adding the balls, occasionally tap the shaft mountings to overcome bearing stiction.
6. Record the number of balls required to raise the block through 90° . This is proportional to the out of balance moment of the block (Wr).
7. Fit an eccentric disc to each block and repeat the above procedure for each block in turn. Enter the results in a table similar to Table 4.1.
8. Remove the extension shaft and replace it in its mounting clip.

Calculation of block positions for balance

Assume that it is required to find the position of blocks (3) and (4) necessary to dynamically balance the shaft, given the positions of blocks (1) and (2). In order to obtain a practical solution for which balance can be demonstrated, it is necessary to take care in choosing the positions of the first two blocks. To assist with this, Table 4.2 gives a number of different configurations which produce practical solutions. The procedure is as follows:

1. Select suitable angular and shaft-wise positions for blocks (1) and (2) by referring to Table 7.2. Note that if the moment values are different, the configurations listed in the table can only be taken as an approximate guide.
2. Find the angular positions of blocks (3) and (4), either by calculation or by drawing (see section 2 and the typical calculations in the next section).
3. Find the shaft-wise displacement of blocks (3) and (4), again by calculation or drawing.
4. Assemble the blocks on the shaft in the given and calculated positions using the slider to set and read the positions. Push the slider to one end so that it is clear of the blocks.
5. Test whether or not the shaft is statically balanced.
6. Replace the drive-belt and the perspex dome, then run the motor and observe whether or not the shaft is dynamically balanced.
7. If the shaft is not balanced, repeat your calculations and check the positions of the blocks to find the mistake.
8. When the shaft has been balanced satisfactorily, move one of the blocks by a small amount. Observe the effect on the balance of the shaft.

Typical Calculations

Suppose that block (1) is placed 18mm from the end of the shaft at zero angular displacement and that block (2) is placed 100 mm from it at 120°. The measured out of balance moments are shown in Table 4.1.

It is first necessary to draw the moment polygon of (Wr) values, as shown in figure 7.2.2 (a), to determine the angular positions of blocks (3) and (4). $(Wr)_1$ and $(Wr)_2$ are known in both magnitude and direction. The vector $(Wr)_1$ has been drawn downwards for convenience. The lengths of vectors $(Wr)_3$ and $(Wr)_4$ are known and arcs can be drawn from the ends of $(Wr)_1$ and $(Wr)_2$ to fix the directions of the unknown vectors.

Figure 7.2.2 (b) shows a graphical interpretation of the angular position of the blocks obtained from the moment polygon. Figure 7.2.2 (c) shows how this is interpreted when positioning the blocks.

The shaft-wise positions of the blocks can be obtained by drawing as shown in figure 7.2.2. Let the unknown distance of blocks (3) and (4) measured from block (1) be a_3 and a_4 , as shown in figure 7.2.2(b). Figure 7.2.2(a) shows the angular positions of the blocks as previously determined. Since block (1) is being used as a reference, it will not appear in the calculations.

Draw a vector (100 x 82) units long parallel to direction $(Wr)_2$. The directions of vectors representing $(Wr)_3$ and $(Wr)_4$ are drawn parallel to $(Wr)_3$ and $(Wr)_4$. Their magnitudes are found by drawing the triangle and measuring the lengths of the sides. Note that the direction of vector $(Wr)_3$ must be reversed compared with the direction $(Wr)_3$. This means that a_3 must be negative and block (3) must be placed before block 1 on the shaft.

From the vector triangle,

$$a_3 = -12mm$$

$$a_4 = 124mm$$

To fit all the blocks on the shaft, block (3) should be positioned 5 mm from the left-hand end. The relative positions of the four blocks are then: -